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A case study on the integration of excess heat from Data Centres in the Stockholm district heating system

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A case study on the integration of excess heat from Data Centres in the Stockholm district heating system

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Abstract

The data centre industry is becoming more and more important due to the rapid increase of digitalisation in our society. However, data centres are large electricity consumers since electricity is needed for both the Information Technology (IT) equipment and the cooling systems, as a certain temperature must be maintained in the server rooms in order to guarantee service operations. Thus, it is important to make data centres less energy intensive and implement a circular economy approach in the sector. One possible way to implement circularity is to reuse the waste heat generated in data centres in district heating networks. However, the potential of using waste heat from low-temperature sources, such as data centres, is mainly unexploited; therefore, more studies are needed in order to inform such use. In particular, it is essential to understand how this potential could be assessed.

The main purpose of this study is to identify the barriers to more heat recovery utilisation and the opportunities that heat recovery contracts can bring from the perspective of key stakeholders linked to DCs' systems operation and service use, such as DC operators, DH operators, and municipalities. The study also aims at understanding how municipalities can enhance a greater integration of waste heat from data centres in district heating systems. To reach the objectives, this study is constructed as an explorative case study on the use of excess heat from data centres in the Stockholm district heating system. Eight stakeholders belonging to those categories were interviewed and the data collected were analysed with a Strengths – Weaknesses – Opportunities and Threats (SWOT) analysis.

The study concludes that the main barriers preventing the implementation of heat recovery investments, in the explorative analysis for the Stockholm county, seem to be more business-related than technical. For example, not having a clear business model in place. In terms of main opportunities, these are related to an improvement in sustainability, such as saving resources by replacing them with excess heat and exploiting an inevitable product instead of wasting it. Moreover, this study also concludes that municipalities can help find suitable places for data centres near the district heating grids both through initiatives like Stockholm Data Parks and city planning.

Key-words: heat recovery, district heating, circular economy, data centres, excess heat.

Sammanfattning

Datacenterbranschen blir allt viktigare på grund av den snabba ökningen av digitaliseringen i vårt samhälle. Datacenter är dock stora elkonsumenter eftersom el behövs både för IT-utrustningen och för kylsystemen, eftersom en viss temperatur måste hållas i serverrummen för att garantera serviceverksamheten. Det är därför viktigt att göra datacenter mindre energikrävande och införa en cirkulär ekonomi inom sektorn. Ett möjligt sätt att genomföra cirkulär ekonomi är att återanvända den spillvärme som genereras i datacentren i fjärrvärmenäten. Potentialen för att använda spillvärme från källor med låg temperatur, t.ex. datacenter, är dock i huvudsak outnyttjad, och därför behövs fler studier för att informera om sådan användning. I synnerhet är det viktigt att förstå hur denna potential kan bedömas.

Huvudsyftet med den här studien är att identifiera hindren för ett ökat utnyttjande av värmeåtervinning och de möjligheter som avtal om värmeåtervinning kan ge ur de viktigaste intressenternas perspektiv när det gäller drift av DC-system och användning av tjänster, t.ex. DC-operatörer, DH-operatörer och kommuner. Studien syftar också till att förstå hur kommunerna kan främja en ökad integrering av spillvärme från datacenter i fjärrvärmesystemen. För att nå målen är denna studie uppbyggd som en explorativ fallstudie om användningen av överskottsvärme från datacenter i Stockholms fjärrvärmesystem. Åtta intressenter som tillhörde dessa kategorier intervjuades och de insamlade uppgifterna analyserades med en SWOT-analys (Strengths - Weaknesses - Opportunities and Threats).

I studien dras slutsatsen att de främsta hindren för investeringar i värmeåtervinning, i den explorativa analysen för Stockholms län, verkar vara mer affärsrelaterade än tekniska. Till exempel att man inte har en tydlig affärsmodell på plats. När det gäller de viktigaste möjligheterna är dessa relaterade till en förbättring av hållbarheten, t.ex. att spara resurser genom att ersätta dem med överskottsvärme och utnyttja en oundviklig produkt i stället för att slösa bort den. I studien dras dessutom slutsatsen att kommunerna kan hjälpa till att hitta lämpliga platser för datacenter nära fjärrvärmenäten, både genom initiativ som Stockholm Data Parks och stadsplanering.

Nyckelord: värmeåtervinning, fjärrvärme, cirkulär ekonomi, datacenter, överskottsvärme.

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List of Abbreviations

CHP	Combined Heat and Power
CRAC	Computer Room Air Conditioner
DC	Data Centre
DH	District Heating
EU	European Union
GHG	GreenHouse Gas
IT	Information Technology
JRC	Joint Research Centre
PUE	Power Usage Effectiveness
SDHA	Swedish District Heating Association
SWOT	Strengths-Weaknesses-Opportunities-Threats

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1 Introduction

The IT sector is expected to consume up to 13% of the global electricity by 2030 (Avgerinou et al., 2017). This value is expected to increase due to our society's higher and higher digitalisation, which can lead to a worrying increment of GHG emissions in the sector depending on the energy sources of future electricity systems (European Commission, 2021). Data centres are the main elements of digitalisation and cloud computing, and their number has increased due to a higher demand for data processing. Their size has also increased (Pärssinen et al., 2019). In a data centre (DC), electricity, besides being used for the IT servers, is also used for the cooling equipment. The heat generated by the IT servers and removed by, e.g. an air conditioning system, is usually discharged to the ambient air. However, this heat could be recovered and reused, thus significantly reducing carbon dioxide emissions (Davies et al., 2016; Werner, 2017a). Waste heat from data centres converted into a heat resource is especially promising in the Nordic countries and could be fed into a district heating (DH) system. However, some challenges prevent the utilisation of this excess heat (Li et al., 2021). Thus, further studies and investigations are needed in order to implement heat recovery strategies from data centres to district heating networks (Oró et al., 2019; Kontu et al., 2019).

In the European Union, increased demand for Data Centres has become a priority. Data centres need to become more sustainable and energy-efficient to become carbon-neutral by 2030 (European Commission, 2021). Sustainability can be enhanced by reusing waste heat and using more renewable energy sources. In order to achieve carbon neutrality, the European Commission relies on legislation such as (European Commission, 2021):

- The Ecodesign Regulation 2019/424 of 15 March 2019 on servers and data storage products (European Commission, 2021)
- The EU Code of Conduct on Data Centre Energy Efficiency (European Commission, 2021)
- The EU Green Public Procurement criteria for data centres (European Commission, 2021)

The objective of the Ecodesign Regulation is to limit the environmental impact of enterprise servers and data storage products by enhancing their energy efficiency and circular economy (European Commission, n.d.). The EU Code of Conduct for data centres is a voluntary initiative that aims at improving energy efficiency and reducing the energy consumption of data centres through best practices. This initiative is managed by the JRC (Joint Research Centre) (European Commission, 2016). Finally, the EU Green Public Procurement criteria for data centres aim at assisting public authorities in procuring EU policy-compliant data centres and services, particularly in terms of environmental impact, energy, and climate change. In particular, the unexploited potential for waste heat reuse from data centres is considered a key environmental aspect and as such, the proposed approach is to ensure waste heat reuse in buildings or district heating networks (European Commission, 2020a). In fact, waste heat reuse from data centres has significant potential since over 98% of the energy consumed in a data centre is dissipated as waste heat and then rejected into the surrounding environment (European Commission, 2020b). However, it might not be feasible to recover waste heat since it is usually low-quality and, for example, there might be a lack of heat demand nearby (Wahlroos et al., 2018). The European Union also aims to increase urban waste heat reuse with projects such as the "ReUseHeat" project which is funded under the Horizon 2020 programme (ReUseHeat, 2017).

Kalmykova et al. (2018) listed waste heat recovery as a circular economy strategy in attempt to provide guidance and further application of a circular economy approach. Also, Andrews et al.

(2021) stated that the data centre sector has to become environmentally sustainable by implementing a circular economy approach. The circular economy principle is to maximise the use and utility of resources, products, and assets and minimise consumption and waste in all forms (European Investment Bank, 2021). Switching from a linear economy (take-make-use-dispose) to a circular one can help reduce environmental and climate impacts and also favour innovation while creating new jobs (European Investment Bank, 2021). In the circular economy approach, the value of products and materials is kept in the loop as much as possible by reusing, repairing, recycling or remanufacturing (Netherlands Enterprise Agency and Holland Circular Hotspot, 2020). Energy cannot be recycled by definition, but it can be cascaded for extended use on lower temperatures (Korhonen et al., 2018). The stakeholders are geographically close at the city level, which can benefit collaboration between the parties involved to close resource loops. However, the application of circular economy in cities has not been investigated enough (Prendeville et al., 2018).

The data centre sector is an energy intensive-sector that has to become more sustainable considering that its energy demand is expected to increase. Waste heat is an inevitable by-product, thus applying a circular economy approach in the sector represents one way to enhance sustainability. By reusing excess heat from data centres in a DH heating system, it is possible to add value to something that would otherwise be wasted. However, some barriers are preventing more utilisation of excess heat from data centres. This study aims to identify how to achieve better integration of excess heat from data centres in a DH system through the perspectives of different stakeholders. To reach that goal, waste heat utilisation in Stockholm was investigated.

1.1 Scope and aim

The scope of this study focuses on the usage of excess heat from data centres in a district heating network, aiming to identify how it could be more integrated considering the stakeholders' perspectives. The three categories of stakeholders identified are DC and DH operators and municipalities. The study is constructed as an explorative case study in the Stockholm county.

1.2 Objectives and research questions

This thesis is a case study on the integration of excess heat from data centres located in the Stockholm county area in the Stockholm district heating (DH) system, which is a well-established network. The cold climate in Sweden is suitable for data centres, considering their need of cooling energy (Wahlroos et al., 2017). According to Data Center Journal (2022), in Stockholm there are 16 providers running 26 data centres. Moreover, heat demand in Stockholm is high due to long and cold winters. Lastly, Stockholm is an important technological hub, home to many start-ups and innovative companies.

This work has two main objectives. Firstly, to identify the barriers and opportunities of using excess heat from data centres in district heating systems. Secondly, to understand how municipalities can enhance a greater integration of heat recovery into a district heating system, considering in particular excess heat from data centres.
Since recovering waste heat from low-temperature urban sources is largely unexploited (RINA-C, 2018), it is important to understand how the use of excess heat from such sources is slowed down and how this waste heat can instead be used successfully.

In order to achieve the objectives, the following research questions are considered:

1. What barriers oppose more utilisation of excess heat from data centres at the urban level?
2. What are the opportunities associated with heat recovery from data centres at the urban level?
3. How can municipalities help to boost heat recovery from data centres?

1.3 Limitations

This work focuses on heat recovery of excess heat from data centres and as such other low temperature urban sources were not prioritised.

A limitation to excess heat recovery techniques was also applied, as this work only considers recovered waste heat to be used in a district heating system. Moreover, reusability and material recovery are not discussed nor are energy efficiency measures that could lessen Data Centres intensive electricity use.

This work is further delimited to specific operators, data centres and district heating companies, and municipalities owning partly the district heating companies involved.

Lastly, it was chosen to have a geographical delimitation by only including operators involved in the Stockholm district heating system. This limitation might make the generalisation to other countries or even cities/regions questionable.

This work was limited by a low response rate from particular municipalities and by the qualitative nature of the SWOT analysis, which offers a start to further studies. Moreover, some economic and business data were considered sensitive and could not be shared with the author.

2 Literature review

2.1 Data Centres and potential for heat recovery

A data centre is a facility aimed at storing data and information, processing and disseminating, organised around a particular body of knowledge or a particular business. In a data centre, there are minor components such as racks, storage devices, switches, and majors such as servers used to analyse, store, and transmit data. The space inside a data centre is usually divided into three zones: IT rooms, IT support areas and ancillary areas. The IT rooms contain equipment and cabling related to computers and telecommunication systems. The IT support areas are power distributing systems, switches, and cooling systems. Finally, the ancillary areas are spaces meant for example for offices (Huang et al., 2020). There are also several different types of data centres, such as co-location, enterprise data centres, and cloud data centres (Andrews et al., 2021). In the particular case of a co-location data centre, multiple customers can rent space and equipment and locate their servers, networks and storage equipment there (European Commission, 2020a). Enterprise data centres are private centres aimed at supporting single organisations, while cloud data centres are owned and managed by companies who rent their virtual infrastructure (Andrews et al., 2021). According to Data Center Knowledge (2021), the largest co-location provider is Equinix with 11.1% market share. An example of a company owning enterprise data centres is Facebook. Finally, cloud services are for example offered by Amazon Web Services (RankRed, 2021).

Data centres are large electricity consumers and require significant amounts of cooling energy. The data centre sector was responsible for 1.3% of the global energy consumption in 2010 (Capozzoli and Primiceri, 2015). The energy consumption of data centres is estimated to increase by 15-20% per year (Capozzoli and Primiceri, 2015). The cooling system can consume up to 40% of the total energy needed in a data centre (Capozzoli and Primiceri, 2015). However, most of the electricity used in a DC is converted into waste heat (Li et al., 2021). Since waste heat is an inevitable by-product, using it in a DH system can contribute to meet the heating demand of end users (Andrés et al., 2018). Moreover, data centres could use district cooling while being connected to a DH system (RINA-C, 2018). This waste heat is usually low quality (<85 °C), which means that it cannot be completely exploited in current DH networks (Wahlroos et al., 2017). Thus, a low-temperature district heating network would enable a higher potential to recover low-quality excess heat from data centres (Wahlroos et al., 2018). Currently, heat pumps, which upgrade the excess heat's quality, can support the use of excess heat from a DC in a DH network (Wahlroos et al., 2018). Potentially, a DC would also be a stable source of heat, considering that it is operating with a similar load profile and power consumption every hour of the day (Wahlroos et al., 2017). However, some barriers limit the utilisation of excess heat from data centres (Wahlroos et al., 2018). Moreover, heat recovery investments need to be economic attractive for both the heat provider (the DC) and the DH operator (Wahlroos et al., 2017).

The energy efficiency of a data centre is usually measured with the PUE (Power Usage Effectiveness) which is defined as the ratio of the total power used divided by the amount of the power used by the IT equipment (Wahlroos et al., 2017). So, a PUE equal to 1 means that all the energy used by a data centre powers the IT equipment (Huang et al., 2020). The PUE is actually overly simplified, but it is still the benchmark metric, thus many data centres do not heat reuse (Wahlroos et al., 2017).

It is questionable if using waste heat from data centres actually reduces CO_2 emissions, as heat pumps are also needed and consume electricity (Wahlroos et al., 2018). So, the environmental impact depends on the emission factor of the electricity used (Wahlroos et al., 2017). The emission reductions for the DH network depend instead on which fuel and technology are replaced by waste heat. For example, if a fossil fuel-based production replaced, waste heat utilisation decreases CO_2 emissions of the DH production (Wahlroos et al., 2017). However, overall making use of waste heat improves the energy efficiency of a data centre, and, as waste heat is an inevitable by-product, its utilisation is an economic, environmental and technical sustainable practice (Wahlroos et al., 2017).

2.1.1 Cooling methods

Data centres need cooling to guarantee efficiency and prevent malfunctions. The optimal air temperature in the server rooms should be between 18 °C and 27 °C (Wahlroos et al., 2018). The most common cooling method used in data centres is to utilise Computer Room Air Conditioners (CRACs) and chilled water. In a data centre, the racks which hold the servers are organised in rows, usually in alternate hot and cold aisles. In the cold aisles, a perforated and raised floor allows the circulation of chilled air. The hot air then exits from the racks into the hot aisles and it is recirculated to the CRAC cooling units. The CRAC cooling units are supplied with chilled water (Wahlroos et al., 2018). In this case, the optimum location for recovering waste heat would be at the air flow returning to the CRAC unit (Davies et al., 2016). The collected waste heat can then be redirected to a heat pump (Wahlroos et al., 2018). In Figure 1 a schematic of a data centre utilising a CRAC cooling unit is shown. Apart from air cooling, other cooling methods include liquid cooling, two-phase cooling, free cooling and district cooling (Wahlroos et al., 2018).

Liquid-cooled systems are promising technologies for cooling data centres with a high power density. Emerging technology is the fully immersed direct liquid-cooled system. In this type of cooling system, the servers are immersed in a dielectric liquid which is used to transfer heat to a water jacket. This heat can then be directly transferred to an external loop and reused. Performance comparisons between this liquid-cooled system and air cooling show that a data centre with a liquid-cooled system has power savings, cooling energy savings, and a lower PUE (Capozzoli and Primiceri, 2015).

The waste heat temperature changes depending on the cooling technology used in a data centre and on the location from which waste heat is taken. In air-cooled data centres, waste heat can be recovered at a temperature between 25 °C and 35 °C (Wahlroos et al., 2017). This heat should be then upgraded using heat pumps (Davies et al., 2016). Waste heat has a higher temperature in liquid-cooled data centres, between 50 °C and 60 °C (Wahlroos et al., 2017). A higher waste heat quality enable more heat recovery opportunities (Capozzoli and Primiceri, 2015).

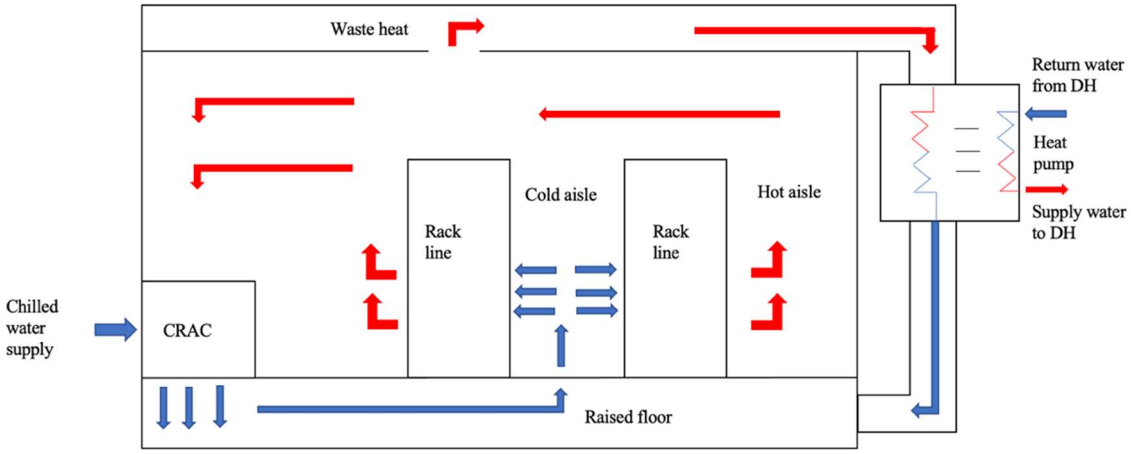


Figure 1 - Schematic of a heat recovery system connected to DH with a CRAC cooling unit (Reproduced from Wahlroos et al., 2018)

2.1.2 Heat recovery in Data Centres using heat pumps

Heat pumps can be used to upgrade the quality of waste heat from a data centre. There are different thermodynamics cycles that could be selected. For example, they could have two or more stages (Davies et al., 2016). Choosing a particular cycle depends on the specific temperature values, the range of temperatures involved and if waste heat is available in one or more streams at different temperatures (Davies et al., 2016). Considering a single-stage cycle, the waste heat initially goes through an evaporator, in which is absorbed by a refrigerant, and then vaporises. The refrigerant vapour then passes through a compressor, which increases its pressure and temperature. The fluid then reaches a condenser, in which it becomes a liquid releasing heat. After the condenser, the liquid refrigerant goes through an expansion valve, reducing its pressure and thus temperature, making it ready again for the evaporator (Davies et al., 2016).
When heat pumps are integrated into a data centre, they can also deliver cooling energy to it by redirecting cooled waste heat to the cooling system. If waste heat is used for a DH network, heat pumps should be connected to both the supply and return side of the DH system (Wahlroos et al., 2017).

2.1.3 Barriers opposing to heat recovery implementation

The main barrier opposing waste heat recovery implementation in data centres is that waste heat is usually low quality (Ebrahimi et al., 2014; Oró et al., 2019; Wahlroos et al., 2018; European Commission, 2020b). Other barriers include (Wahlroos et al., 2018) the lack of heat demand nearby, high investment costs, the different business logic of DC and DH operators, information security and reliability and business models.

For heat recovery investments to be feasible, the DH network should be nearby the data centre otherwise, connection fees would be too expensive as those fees increase with the distance. Heat recovery equipment can also be expensive, even though the payback period could only be a few years (Wahlroos et al., 2018). Regarding the different business logic, the DC operator business logic is investment-intensive and ownership is usually made by private investors. The DH operator business logic is also investment-intensive, but ownership is both private and public, and there are natural monopolies inside the networks. The rate of return of a DC operator is significantly higher than the one of a DH operator. Different expected rate of returns might create unbalanced expectations between DC and DH operators (Wahlroos et al., 2018).

Information security is critical for DC operators. Their markets are highly competitive, thus DC operators limit the data available. The lack of data and transparency can contribute to lessening the potential of commercialising excess heat (Wahlroos et al., 2018). Business models for heat recovery are usually missing or not clear (Wahlroos et al., 2018). RINA-C (2018) also argues that clear business models have to be applied in order to successfully implement heat recovery contracts, considering that DC and DH operators have different expectations. Moreover, the installation of waste heat recovery equipment should have a minimum impact as data centres operators need to have information security and continuous operation guaranteed (RINA-C, 2018). It is also important to consider that data centres produce more waste heat when more energy cooling is needed, making waste heat more available during the summertime when there is less or no demand from DH. However, thermal storage could also be used in order to provide waste heat when the demand is higher (Wahlroos et al., 2018).

The barriers that minimise waste heat utilisation are mostly due to a lack of proper investments and incentives rather than technical problems (Wahlroos et al., 2018). Koronen et al. (2020) also argue that the development of policy incentives for waste heat recovery is a key phase for having more excess heat reuse and that there should be a greater effort in developing future policies. Moreover, some countries provide a lower electricity tax to data centres, considering them "energy-intensive industries", in order to attract DC operators, but this is done without demanding anything in return (Korenen et al., 2020). As a result, the financial incentives for energy efficiency are weakened and the geographical siting incentives are enhanced (Korenen et al., 2020).

2.1.4 Investments in heat recovery technology

The commercialisation of waste heat from data centres has barely been done, even though studies in the literature show that there are significant energy cost savings and low payback periods (Wahlroos et al., 2018). Its utilisation requires investments in heat pumps, piping, and connection to a DH network. Moreover, heat pumps use electricity which increases operational costs. Heat pumps also have operation and maintenance costs. However, selling waste heat also brings revenue to data centres (Wahlroos et al., 2018). Nonetheless, the business models for this kind of heat recovery are currently not clear. The key question which arises for both the DC operator and the DH operator is who pays for the heat recovery equipment. Moreover, the negotiation process between them should be as transparent as possible, since they reason with a different business logic. Generally, the first priorities for commercialising waste heat are profitability and quality of the heat. For example, if the DC operator pays for the equipment, the revenue from waste heat must cover the costs needed for the investment in a reasonable time period. From the point of view of the DH operator, waste heat should be less expensive than the marginal production costs of the replaced heat (Wahlroos et al., 2018).

Moreover, data centres usually have contracts that last between 1 and 5 years, but heat production unit investments have a technical lifetime of 15-40 years (Wahlroos et al., 2017). However, data centres may be unwilling to have such long-term contracts (Wahlroos et al., 2017). For the DH operator, buying waste heat could mean lower production costs as waste heat would replace more expensive peak load production (Wahlroos et al., 2017). DC operators could increase their green image (RINA-C, 2018).

2.1.5 Existing applications

According to RINA-C (2018), there are not many cases of data centres supplying their excess heat to a DH system. There is a ReUseHeat demonstration in Brunswick, Germany, which is listed as a best practice in the ReUseHeat project. Other case studies can be found in Denmark, where Apple, Google, and Facebook are establishing large data centres and they intend to supply their waste heat to a DH heating system. However, large data centres are usually far from cities and DH grids thus the utilisation of waste heat might not be feasible (RINA-C, 2018). Other applications can be found in Finland. Tieto DC in Espoo supplies the DH network with 30 GWh of waste heat annually (Wahlroos et al., 2018). TelecityGroup heats 4500 block apartments and 500 detached houses with three data centres in Helsinki (Wahlroos et al., 2018). It has to be noted that many DC operators do not reveal much information on heat recovery and data on the actual reused waste heat (Wahlroos et al., 2018).

2.2 District heating

District heating distributes heat by using an underground network of pipes, which circulates either hot water or low-pressure steam from centralised plants and heat-producing units to residential, commercial, and industrial buildings. The energy sources for district heating vary and include fossil fuels, waste incineration, waste heat, and biomass. District heat can be produced through combined heat and power (CHP) plants, generating electricity and heat. CHP plants can enhance energy efficiency and reduce both greenhouse gas (GHG) emissions and fuel consumption (Rezaie and Rosen, 2012).

In a district heating system, there are three main subsystems (Rezaie and Rosen, 2012):

1. Source of thermal energy
2. Distribution network
3. End users

The first element provides heat in the form of steam or hot water, which can be produced from heat plants or CHP plants. In these plants fuels such as natural gas and biomass are combusted. Other units generate heat as geothermal plants, solar collectors, and heat pumps. The distribution network consists in a network of underground pipes in which the heat carrier circulates till the end users and then returns to the source through the return pipes. In this piping network, heat losses can be critical. The end users consists of all buildings requiring space heating and domestic hot water (Rezaie and Rosen, 2012). The heat is distributed inside buildings through distribution pumps, which require electricity (Werner, 2017b).

Historically, there are four generations of district heating systems (Lund et al., 2014). The first generation used steam as a heat carrier and was introduced in the U.S. in the 1880s to remove individual boilers in apartment buildings. This system is now considered outdated as using steam generates more heat losses and is more dangerous for pedestrians due to possible steam explosions. The second generation used pressurised hot water as a heat carrier, with supply temperatures over 100 °C. The introduction of these systems happened in the 1930s and dominated district heating systems until the 1970s. This second generation was mostly introduced for obtaining fuel savings and using CHP plants. The third and current used generation, introduced in the 1970s and boomed in the 1980s, still makes use of pressurised water, but the supply temperatures are lower than 100 °C. This generation was born due to two oil crises, which led to the substitution of oil with coal, biomass and waste. The fourth generation of district heating, for which Lund et al. (2014) indicates 2020 as starting date,

should permit lower distribution temperatures (Lund et al., 2014) which improve efficiency and decrease the supply cost (Averfalk and Werner, 2020). In particular, the temperature level of the fourth generation of DH is set to be as close as possible to the actual temperature demand of end-users, that is around 60-70 °C maximum. This feature would decrease grid losses and better integrate more waste heat sources such as excess heat from data centres and supermarkets. The fourth generation will also be very compatible with low-energy buildings (Lund et al., 2021). Per definition, the fourth generation of DH should be able to fulfil five abilities (Lund et al., 2021):

1. The ability to supply existing, renovated and new buildings with low-temperature DH for both space heating and domestic hot water
2. The ability to distribute heat with low grid losses
3. The ability to use waste heat from low heat sources and integrate renewables
4. The ability to be an integrated part of smart energy systems
5. The ability to ensure suitable planning, cost, and incentive structures

Several studies also mention an additional generation of district heating, that is the fifth generation of district heating and cooling. In this generation, grid losses are minimised by having ultra-low temperature heat that requires an upgrade at the end-users with heat pumps. Such a low distribution temperature would boost even more the exploitation of waste heat from low-temperature waste heat sources by enabling a direct heat injection to the system. However, Lund et al. (2021) came to the conclusion that the fifth generation should not be seen as a sequential development of the fourth but rather as a parallel development thus, they will not substitute each other but will be complementary options (Lund et al., 2021).

Concerning the costs of a district heating system, the total sum depends on three major factors (Rezaie and Rosen, 2012):

1. The production cost of the heat
2. The costs associated with the piping distribution network
3. The end users' connection costs

In particular, the total costs of a district heating system are mostly affected by the costs of the pipe distribution network, which mainly depend on the pipe length and diameter. Reducing these costs, together with heat losses and the pumps' electrical energy consumption, would improve the economic competitiveness of the district heating system (Pirouti et al., 2013). For instance, in order to keep the distribution costs low, short distribution pipes are favourable (Werner, 2017a). A district heating system is thus more feasible where heat densities are high (Averfalk et al., 2017).

The heating markets can be regulated or deregulated. There is no competition between district heating and other heat options in regulated markets, whereas in deregulated markets, district heating competes with other heat options (Zhang et al., 2013). The method for product pricing also changes depending on if the heating market is regulated or not. In regulated heating markets, a cost-plus method is usually applied in which all the costs associated with heating are added to what is charged with a profit margin to get the final price. One of the consequences of having this type of pricing method is that efficient companies that manage to reduce their costs are punished with lower profits. Thus, companies are not incentivised to find cheaper and more efficient ways of producing heat. In deregulated markets, the method usually used is the marginal-cost method. However, this method is based on the ideal market theory and is difficult to achieve, especially in natural monopoly markets such as heating markets. Marginal-cost pricing could be approximately realised with unregulated market bidding like in Sweden, where district heating companies are free to set prices (Zhang et al., 2013). However, not having

regulation in district heating systems means that end users might not be protected from possible market power abuse by district heating companies since they cannot choose between district heating providers. That is because district heating companies are not in competition with each other but have a monopoly over their market (Wissner, 2014).

District heating is considered to be environmentally friendly since the emissions in heat sinks, like in residential areas, are avoided by generating heat in central plants. Moreover, using CHP plants is a more efficient way to reduce CO_2 emissions than having plants with heat and electricity production separated. Using more and more renewable energy in district heating systems also boosts sustainability and help combat climate change (Wissner, 2014). However, district heating is often neglected due to a lack of suitable tools needed to analyse it, so the benefits it could bring are often overlooked (Connolly et al., 2014).

2.2.1 Waste heat as a heat source

Waste heat recovery could boost a system's efficiency by using heat that would otherwise be wasted (Arnaudo et al., 2021b). Moreover, using excess heat could imply a lower energy demand, decreased CO_2 emissions, and economic gains. The heat recovered could be either used for internal applications, such as in industrial processes, or for external uses, to generate electricity or to be fed into a district heating network. The integration into a district heating system can be direct or after upgrading the excess heat temperature (Broberg Viklund and Karlsson, 2015).

Industrial excess heat can be used as a heat source for heat pumps. However, industrial heat recovery investments face many challenges. For example, technical solutions which are more complex than expected and different perspectives on heat quality. Moreover, as an additional barrier, there is the risk that the industrial company providing excess heat could shut down, relocate or modify the processes which produce excess heat. However, excess heat recovery investments could actually reduce said risk, considering that they provide additional revenue for the industrial company (Lygnerud and Werner, 2018). Another relevant barrier is that excess heat is usually available in different sections of an industrial process, at different temperatures. Having distributed heat sources would require small-scale setups for heat recovery, which are generally less efficient and more expensive than large-scale setups (Xu et al., 2019).

In Sweden, the share of recovered industrial excess heat is high compared to other countries, even if it was only 9% on the total heat supply in 2014 (Lygnerud and Werner, 2018). The first heat recovery investment in Sweden for district heating was carried out in Helsingborg in 1974. This early development can be explained by introducing taxes on oil in the 1980s and a CO_2 tax on fossil fuels in 1991. However, policies supporting biomass and waste CHP plants have put industrial excess heat recovery in competition with these plants. Despite the long Swedish history of heat recovery, its potential is much higher than the actual recovery (Lygnerud and Werner, 2018).

Other main heat sources for heat pumps are treated water sewage, ambient water, and heat recovered from data centres. Heat pumps have low specific investment and installation costs and can boost district heating cost-efficiency by operating when the electricity prices are low and shutting down when they are high. However, the environmental benefits of heat pumps depend on how the electricity input is generated (Averfalk et al., 2017).

Considering an urban context, several possible waste heat recovery sources exist, including data centres, supermarkets, and ice rinks. The recovery from these sources could be increased by adopting a low-temperature district heating network, with which a larger quantity of heat can be captured without a temperature boost. Moreover, lower temperatures generate fewer heat losses during the distribution phase (Arnaudo et al., 2021b). However, it could be challenging to recover low-temperature waste heat, as its recovery is, for example, affected by the user's demand, limited space for heat recovery facilities, and payback period (Xu et al., 2019).

2.3 Overview of the Stockholm case study

2.3.1 District heating

The first Swedish district heating system was introduced in Karlstad in 1948 when the heat generated in a CHP plant was provided to an industrial facility. All main cities in Sweden have a district heating system (Werner, 2017b). Four major driving forces were identified that brought the introduction of district heating systems (Werner, 2017b):

1. The municipal interest in the CHP plants in the 1950s and 1960s
2. The million homes programme during the period 1965-1974
3. The two oil crises in the 1970s
4. The several climate change policies introduced in the 1990s

In 1996, with the deregulation of the national electricity market, non-municipal owners acquired many municipal district heating systems (Werner, 2017b). However, the deregulation of district heating was not investigated as much as the deregulation of the electricity market, setting only as guiding principle "a business-like manner" principle (affärsmässighet in Swedish). This new legislation has led district heating companies to switch from cost-based pricing to price-setting. Thus, considering the weak competition in the district heating market, price regulation has been frequently contemplated. In response to that matter, the Swedish District Heating Association (SDHA) has argued that price regulation is unnecessary since district heating companies operate on a heating market and not on a district heating market. As such, they compete with other technologies like heat pumps. Currently, there is a voluntary price controlling system issued in 2013, but it is still unclear if it is enough to protect customers from over-pricing (Åberg et al., 2016).

Stockholm has had district heating since the 1950s (Levihn, 2017), which now has more than 80% of its buildings connected to it (Su et al., 2021) and consists of a third-generation technology (Arnaudo et al., 2021a). District heat is mostly produced at four main plants in Värtan, Högdalen, Hässelby and Hammarby. The total length of the pipe distribution network is around 2800 km, and annually the district heat delivered is around 6000 GWh within the geographical area of the City of Stockholm (Su et al., 2021). District heating is mainly supplied by waste incineration, biomass CHP plants, and heat pumps utilising heat from treated sewage water, district cooling, and seawater (Levihn, 2017). Fossil fuels are still used during extremely cold days for peak heating (Su et al., 2021). There are four main actors involved in the Stockholm County district heating systems (Levihn, 2017): Stockholm Exergi (previously called Fortum Värme), which is the biggest one, Söderenergi, Norrenergi, and EOn.

Each of these companies owns its part of the network in which they also have a monopoly (Levihn, 2017). Stockholm Exergi is the utility that produces district heating and cooling for Stockholm, and it is co-owned by Fortum and the City of Stockholm (Fortum, n.d.). Its network also supports third-party access under the name of "Open District Heating" (Levihn, 2017), in

which companies can sell their excess heat at market prices, thus enhancing heat recovery (Open District Heating, n.d. a).

The general supply temperature inside Stockholm Exergi's network is 68 °C, with a return temperature between 35-55 °C (Levihn, 2017). Stockholm Exergi also produces 0.45 TWh of district cooling annually, using heat pumps and free cooling through sea water (Levihn, 2017). Söderenergi produces district heat for the region of Stockholm, in particular for Södertälje, Nykvarn, Huddinge, Botkyrka, and Salem (Söderenergi, 2019). Norrenergi provides district heating and cooling for the northern part of Greater Stockholm, particularly for Solna, Sundbyberg, Bromma, and Danderyd (Norrenergi, n.d.). Finally, EOn also produces district heat for the region of Stockholm (EOn, 2021).

In the future, the Swedish district heating systems will have as challenge lower heat demands since buildings will have to comply with the near-zero energy requirements stated in the European energy performance of buildings directive (Werner, 2017b). Said challenge consists of maintaining the economic feasibility, which could be done by reducing distribution losses with better pipe insulation and lower distribution temperatures. Moreover, the economic feasibility could also be reduced if many customers decide to switch from district heating to an alternative like ground source heat pumps, as fewer customers would carry the fixed sunk costs of the network. These sunk costs usually cover the investments put into plants and grids (Åberg et al., 2020).

2.3.1.1 Open District Heating

Businesses and companies that generate excess heat and are close to Stockholm Exergi's district heating network can sell their excess heat under the third party access "Open District Heating". Stockholm Exergi invests in the piping network needed to connect the suppliers with their district heating network and the suppliers in the necessary production plants, like heat pumps. According to standard models based on projected outdoor temperatures, suppliers are paid monthly depending on the delivery temperature and energy volumes (Open District Heating, n.d. b). There are also two different contract models. The first model is called "Open District Heating Call", with which suppliers agree with Stockholm Exergi on what levels of capacity should be supplied to the district heating system. In this model, Stockholm Exergi demands heat when it is needed, guarantying that these requests are only made when the outdoor temperature is 12 °C or below and the suppliers are paid with a fixed monthly payment for capacity delivered and with a variable energy payment for delivered heat. This model is suitable for data centres because they have stable levels of excess heat throughout the day and year. The second model is called "Open District Heating Spot", with which suppliers decide when to deliver excess heat and no specific capacity demands are made. Then, suppliers are paid based on how much excess heat they deliver. These payments depend on the outdoor temperature and are higher when the outdoor temperature is lower (Open District Heating, n.d. c).

2.3.2 Heat recovery from Data Centres in Stockholm

Since 2008, more and more technology companies decided to build their data centres in Sweden (Libertson et al., 2021). That is due to different reasons. In Sweden, the electricity price is one of the lowest in Europe. Moreover, this price is reduced for data centres over 5 MW (Rankinen et al., 2021). The Swedish electricity grid has a high reliability, of about 98.7% (Rankinen et al., 2021), which is an important factor for DC operators since outages are expensive for them (Rankinen et al., 2021). Sweden has also a high share of renewables in its electricity production

(Rankinen et al., 2021). In Sweden there are also important fibre optical companies, that enable a fast communication. The Swedish climate also favour DC operators since a cold climate translate to a less expensive cooling system. Moreover, in Sweden there is also a low risk of natural disasters such as earthquakes and floods (Rankinen et al., 2021).

In Stockholm, there are several applications of heat recovery from data centres. An example is offered by Bahnhof Thule data centre that heats 1050 apartments with its excess heat (Stockholm Data Parks, 2017).

Stockholm Data Parks is an initiative by Stockholm municipality, Invest Stockholm, Stockholm Exergi, power grid operator Ellevio, and dark fibre provider Stokab (Stockholm Data Parks, 2021). It aims to prepare and offer the necessary infrastructure for building data centres in three particular sites (Stockholm Data Parks, 2021):

1. Data Park Kista
2. Data Park Brista
3. Data Park Skarpnäck

The data centres built on those sites are connected to the district heating network in order to recover waste heat (Stockholm Data Parks, 2021). In particular, the data centres are co-located close to a central heat pump station that is needed to upgrade the waste heat quality, and then the upgraded heat can then be fed to the district heating network. This way, the district heating operator does not rely on a single data centre but on a combination of them (Koronen et al., 2020). The long-term objective is to supply 10% of the heating demand through excess heat recovered from data centres, with the aim of helping Stockholm city to be fossil-free by 2040 (Invest Stockholm, n.d.).

Vattenfall and Cloud&Heat Technologies have also launched a pilot project in Jordbro, where a new data centre was built next to Vattenfall's biomass-fired heating plant. This project aims to utilise the excess heat from the data centre in the DH plant (Vattenfall, 2021).

3 Methodology

3.1 Research design

This study is designed as an exploratory case study using a qualitative approach. An exploratory approach was selected as it is well-suited to investigate a specific area and lays the foundation for future research (Kumar, 2011). Moreover, case studies are useful for exploratory research as they are suited to new research areas (Rowley, 2002). Case studies are also effective when there is a need to gain insights into specific topics (Krusenvik, 2016).

Interviews were chosen as a data collection method since this study aims to collect qualitative data and obtain insights from key stakeholders (Rowley, 2012).

Lastly, the interview responses were analysed with a SWOT analysis. A SWOT analysis was considered an appropriate method for answering the research questions since it is useful when there is a need for profiling and individuating issues (Helms and Nixon, 2010). SWOT is also a main strategic tool for listing strengths and weaknesses (Coman and Ronen, 2009).

3.2 Research process

Firstly performed was a literature review, which provided an understanding of data centres, their functioning and energy needs, as well as heat recovery potential. Also investigated were district heating and heat recovery. The literature review served as a base on which to build interview questions. The scope of the literature review was narrowed down to excess heat from data centres to be used in a district heating network but started broader with a general focus on heat recovery. This thesis was developed as a case study set in Stockholm, and information on Swedish district heating systems was also gathered.

The second step consisted in individuating the stakeholders involved in the case study. Then, the data collection began by doing semi-structured interviews with the selected stakeholders. The questions asked were open-ended in order to collect qualitative data. The final step involved a SWOT analysis made of the collected data. Figure 2 shows the research process graphically.

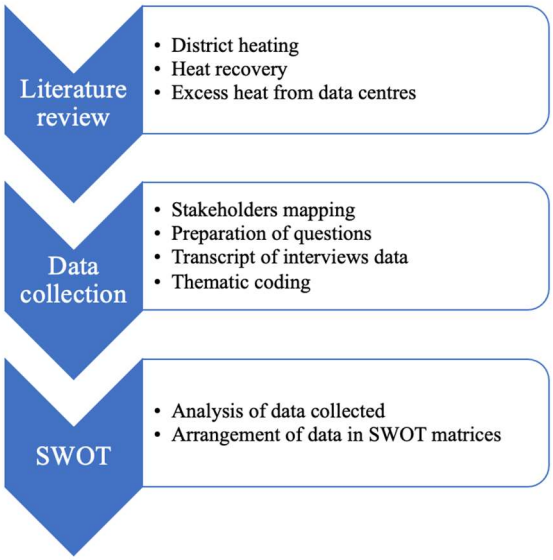


Figure 2 - Research process

3.3 Literature review

Relevant literature is essential for every type of research project (Snyder, 2019). It provides a state of the art on the research topic and helps identify research gaps that can lead to future work (Paul and Criado, 2020).

Firstly, I performed a broad literature review on the chosen energy system, district heating, and excess heat recovered in a district heating network. This step was necessary to acquire knowledge on the topic. The search on excess heat recovery was at first focused on different heat resources but was then narrowed down to only data centres. After realising its novelty and potential, I chose to consider only data centres as a possible heat source for district heating networks. The literature review helped shape the objectives and research questions of this study. Since this study is an explorative case study, it was necessary to gather information on the Stockholm district heating network through academic literature and grey literature sources. The interview questions were based on previous literature. Google Scholar was chosen instead of Scopus because it allows to gather grey literature as well and recent and highly cited papers were prioritised. The keywords used to fulfil the search are *district heating, heating market, heat recovery, data centres, excess heat reuse, Stockholm.*

3.4 Data collection

An interview study was conducted in order to gather knowledge from key stakeholders. The interviews were semi-structured, with open-ended questions allowing interviewees to express themselves freely, removing the possibility of investigator bias (Kumar, 2011).

A total of 20 stakeholders were contacted using the contact details found on the companies' websites, municipalities' websites, and LinkedIn. Out of these, eight agreed to do an interview, three declined and the remainder did not respond.

The interviews were conducted through Zoom between 25/01 and 01/04 of year 2022, and recorded with the interviewees' permission, except for one stakeholder who preferred to use Microsoft Teams and did not want the interview to be recorded. The recordings were only needed for the transcript of the data phase and will not be kept by the author. The interviewees were kept anonymous to guarantee privacy and safety of data and reminded before every interview of the possibility of declining to answer specific questions. Also, the interviewees were previously provided with the respective interview protocol. The interviews were held in English.

3.4.1 Key stakeholders

The first step of the data collection was selecting the key stakeholders involved in data centres and district heating in Stockholm. The stakeholders identified for this study were divided into three main categories:

1. DH operators
2. Municipalities
3. DC operators

The DH operators were found while conducting the literature review by using both academic literature and the companies' web pages. The ones contacted were selected considering that they serve the county of Stockholm and that they are co-owned by a municipality. The

municipalities contacted are those which partly own the district heating companies selected. The municipalities were identified through the district heating companies' web pages. The DC operators contacted have their data centres localised either in one of the sites of Stockholm Data Parks or outside of them but still in Stockholm county.

Table 1 shows the stakeholders who accepted to have an interview, divided by each category. As can be seen from the table, I assigned to each stakeholder a letter in order to guarantee anonymity. Thus, in this work the stakeholders are referred to as *DH operator A* or *DC operator G*, for example. The order of the letters assigned is only alphabetical and has no other meaning.

Table 1 - Stakeholders interviewed

Categories		
DH operators	Muncipalities	DC operators
DH operator A	Municipality C	DC operator D (in Stockholm Data Parks)
DH operator B		DC operator E (in Stockholm Data Parks)
		DC operator F
		DC operator G
		DC operator H

DC operators D and E were found on the Stockholm Data Parks webpage.

The other DC operators were found through a Google search, considering the question initially: "Which are the biggest data centres in Stockholm?". Excluding the biggest ones, the others were found by simply considering Stockholm county as their location.

Figure 3 shows how each category is involved in heat recovery investments, if directly or indirectly, and their correlations between each other. DH and DC operators are considered to be directly involved as they are the actors doing a contract in order to commercialise waste heat. Municipalities are instead considered to be indirectly involved as they partly own DH operators, but do not participate in a heat recovery contract. However, as can be seen from Figure 3, they could boost heat recovery investments through initiatives, as Stockholm Data Parks, incentives and policies, for example.

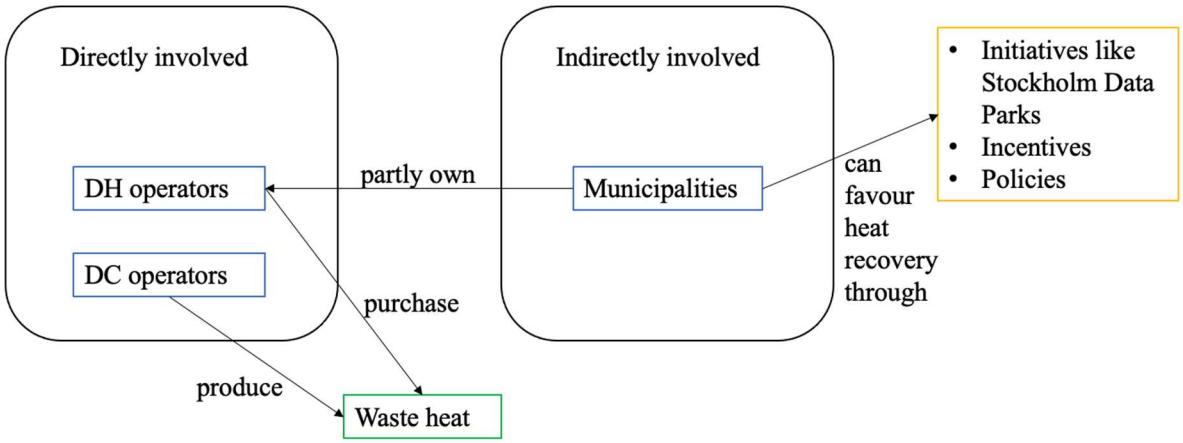


Figure 3 - Rapresentation of stakeholders' categories involved

3.4.2 Interview protocols

Four different interview protocols were developed in order to adequately address each category of stakeholders identified. Elaborating a focused protocol for each category was an obligatory step as the stakeholders individuated in this study differ from each other. For instance, they are involved in heat recovery investments differently and play different roles. The protocol for municipalities differs a little more in particular, as municipalities are not directly involved in heat recovery investments. Moreover, the municipalities' protocol was particularly designed to answer the third research question. Sometimes, the protocols were slightly adjusted to a specific stakeholder prior to the interview to better fit the interviewees' profiles. However, the protocols, except for the one meant for municipalities, were written considering four main themes, including an introduction and wrap up part:

1. Introduction
2. Economic aspects
3. Technical aspects
4. Business aspects
5. Environmental aspects
6. Wrap up

The introduction questions aim to obtain general knowledge about the interviewee and their company, except for municipalities. The introduction section in the municipalities' protocol serves to know how much the specific municipality is aware of heat recovery from data centres and how much they are involved in their district heating network's business. This section's last question is important to understand how much municipalities can actually influence district heating companies' decisions. The wrap-up section should give the interviewee a space to express everything they feel important on matters that weren't specifically asked. Moreover, in those last sections, there are also questions that didn't belong in the previous ones but were considered relevant to be asked.

Regarding the specific themes chosen, I decided to focus my questions on economic, technical, business and environmental aspects since they are mostly covered in the literature reviewed, on which the questions developed are based. Moreover, I consider these topics the most relevant ones for this study. Finally, the protocols were designed with regard to the objectives and research questions of this study. The entire protocol can be found in the Appendix.

3.4.3 Transcript and thematic coding of interviews data

The interviews conducted through Zoom were transcribed verbatim while listening to the recordings. This was a necessary step in order to be sure of not losing important data. Transcription of data is also important for becoming familiar with the key points and perspectives raised and for taking the time to fully understand what has been said by the interviewee (Rowley, 2012). Regarding the interview done with Teams, I took notes while interviewing the stakeholder and later organised my notes in the same transcript format.

The transcripts were then carefully read and thematic coding was done. The selected themes were the four ones individuated for the interview protocols. Practically, the data from the interviews were sorted and assembled under each theme as to compare the answers and different perspectives of the stakeholders. This step prepared the base for the SWOT analysis.

3.5 Data analysis

SWOT (Strengths, Weaknesses, Opportunities, Threats) analysis is a strategic planning tool used to evaluate four variables, divided by internal and external, usually presented in a matrix. The internal variables are strengths and weaknesses and the external opportunities and threats (Demont and Rutsaert, 2017).

Besides being one of the most popular tools used in strategy and decision planning, SWOT is not exempt from criticisms due to not having its variables hierarchically ranked and being a one-time shot of a precise moment in time (Coman and Ronen, 2009). Moreover, SWOT variables also lack a quantitative weighting (Helms and Nixon, 2010).

However, SWOT offers a helpful basic framework to list issues and opportunities and can be used as a first step for strategic planning and policy implementation. Thus, it is appropriate to fulfil the research questions of this work. Finally, Helms and Nixon (2010) argue that linking SWOT with expert interviews should be considered for the future development of this analysis, which has been done.

In this study, a SWOT matrix was done for each of the main topics individuated in the interview protocols that is economic, technical, business, and environmental aspects. This was considered the best approach to analyse the data collected for clarity reasons and consider that each topic is different and needs to be evaluated per se. The barriers opposing to more heat recovery contracts were identified considering the variables “weaknesses” and “threats”, whereas the opportunities considering “strengths” and “opportunities”.

4 Results and discussion

This chapter begins with an overview of the stakeholders interviewed. Then, SWOT analysis follows. The obtained data is then compared to the literature review. Finally, the chapter ends with a discussion on social responsibility and stakeholders' roles.

4.1 Overview of key stakeholders

In Table 2, relevant details of the stakeholders interviewed are presented.

Table 2 - Details of the stakeholders interviewed

	Does it buy/sell excess heat?	Is it in Stockholm Data Parks?	Does its co-owned DH utility buy excess heat?
DH operator A	Yes	/	/
DH operator B	No	/	/
Municipality C	/	/	Yes
DC operator D	Soon	Yes	/
DC operator E	Yes	Yes	/
DC operator F	Soon	No	/
DC operator G	Yes	No	/
DC operator H	Yes	No	/

Considering the stakeholders involved directly in a heat recovery investment, DH operator A stated that they buy excess heat from third parties as data centres and DH operator B that they do not since there are no data centres available near their heating grid. They, however, considered often excess heat from different production industries. DC operator D is currently building their data centre in one of Stockholm Data Parks. DC operator E has six data centres in one of Stockholm Data Parks. DC operator F is in the process of building the necessary heat recovery equipment in both of their sites and will be selling their excess heat soon. DC operator G sells its excess heat from multiple locations. DC operator H sells its excess heat from two different sites, but only one of them is in Stockholm. Regarding municipalities, Municipality C owns 50% of the DH operator in its city and has political representation on the board. Municipality C promotes reusing waste heat and helps city planners find good locations for data centres where they can be connected to the DH system and deliver their waste heat.

4.2 SWOT analysis

A SWOT matrix was performed for each of the four thematic areas individuated in the interview protocols. In order to develop the SWOT matrices, the data gathered were analysed considering the four variables the SWOT analysis is made of:

1. Strengths
2. Weaknesses
3. Opportunities
4. Threats

Previous to each SWOT matrix, the data used for the analysis is presented. SWOT is employed to help answer the research questions, which are answered in chapter five.

4.2.1 Economic aspects

- Perspectives of DC operators:

Overall, a heat recovery investment does not seem to be economically attractive. DC operators F and G stated that they had to invest in heat pumps to upgrade the excess heat's quality from 25-28 °C to around 70 °C, which is an additional investment for them. Moreover, heat pumps consume electricity which is also another cost. Other costs are very dependent on the type of data centre, but for example, DC operator F being a co-location data centre had also to invest in redundancy systems. DC operator H broke down their costs into four different components, each generating additional costs that a DC operator that does not do heat recovery does not have. The first cost component is associated with the design phase, as there is more design work to figure out the best approach for recovering the waste heat and upgrading its quality. A control system that works with all the different pieces and components is also needed. The second cost component is CAPEX since everything designed to run a data centre with heat recovery must be bought. The third cost component is OPEX, which includes the cost of electricity needed to run the heat pumps. The fourth cost component is the maintenance cost, which is higher in a data centre with heat reuse since there are more components to take care of, especially the heat pumps. However, even though economically selling excess heat is not that good of a deal, DC operators F, G and H agree that it is going to be profitable in the long run. DC operator G also said that designing a data centre with heat recovery from the start can help save money instead of adding it later on. That is because a data centre designed to recover waste heat can have just one smaller free cooling system as backup and district cooling as the primary system instead of having two free cooling systems.

All the data centre operators interviewed agree that the motivation behind selling their excess heat is not commercial but environmental and that recovering their excess heat is a social responsibility. In particular, DC operator D stated that: "*In general, I would say the data centres I know that heat reuse, the goal is not to make money from heat reuse, the goal is to make a cost-neutral. Everybody wants to do the right thing and reuse heat, we do not want to burn the excess to the atmosphere and we accept that there is a cost to it, we just want to minimise the extra cost really. Saying that we go into this to make money would be completely wrong, that is not the situation in Stockholm.*" DC operator D also declared that being in Stockholm Data Parks has helped them compensate for the extra costs. DC operator H mentioned that they do not make a big profit from their heat recovery investment, but it adds sustainability. More and more customers also want to deal with greener data centres, so a heat recovery investment plays an important role marketing-wise as well, as stated by DC operator F with "*The motivation behind is not commercial as it is not that good of a deal to basically just look at it as heat recovery investment. It is more of a bigger perspective, what kind of role do we play in the society as data centres and where we are located and the amount of energy that we use. So it is more of being aligned with society and also from a marketing perspective of course.*".

- Perspectives of DH operators:

Concerning the DH operators, DH operator A explained that they usually buy and use excess heat from data centres, but since they prefer to use their existing heat pumps, sometimes the electricity price can be so high that excess heat does not have a value for them as it would be too expensive to produce it as primary energy. DC operator H also mentioned that if they had to pay the electricity too much, they would need more compensation for their excess heat but

then a DH operator would not buy their excess heat as it would be too expensive. But DC operator H said that they have a fixed price for electricity.

DH operator A mentioned that there could be a competition with other heat sources as waste, which can have a negative fuel cost. However, DH operator A also said that having more data centres near their grids could be an important opportunity to lower their production costs, even though it can also be challenging to get data centres close to the city since space is expensive in Stockholm.

DH operator B stated that their heat production is cheap hence there could be the possibility of not being able to buy excess heat from data centres even with data centres nearby their grids. That is because the excess heat from data centres would need to be so cheap that it would not have any business value. Also, the potential use of excess heat depends on how the heating grid looks and what production units are in it. Moreover, DH operator B mentioned that selling and buying excess heat could be more important in marketing than in finance for data centres and district heating companies.

- Perspectives of municipalities:

The interviewee of Municipality C does not think that financial aid and policies are needed since a marketplace is already established, which is the most important incentive for data centre operators.

The SWOT matrix regarding the economic aspects can be seen in Figure 4.



Figure 4 - SWOT matrix for the economic aspects

4.2.2 Technical aspects

- Perspectives of DC operators:

According to DC operator F, introducing another technology can bring some technical concerns as in their case they are dependent on selling capacity to their customers who control how much usage they put in the data centres so selecting the size and model of the heat pump can be a concern. But DC operator F thinks that: "*It is important that we take those challenges and be part of doing as much as possible in the community we are located in.*". Also DC operator H mentioned that retrofitting an existing data centre with heat recovery could be challenging. However, DC operator F stated that one of the reasons, if not the biggest, that they are working to connect their data centres to the district heating system is the possibility of making a data centre more sustainable by recovering the excess heat and that in Stockholm the opportunity to recover it is significant. DC operator E also agrees that Stockholm offers excellent opportunities to sell excess heat and add value to it.

DC operator D mentioned that designing the cooling system in a heat reuse data centre is technically more complicated than a traditional cooling system, which is one of the reasons why some data centre operators avoid recovering their excess heat. In particular, heat pumps need to be put in series but data centres work with parallel systems and there also are more complicated control systems. DC operator D also had to install a local cooling system in case the DH operator has a malfunction in their plants. According to DC operator D designing the cooling system was indeed the biggest challenge, as it is an iterative process, and compared to traditional cooling systems there are fewer suppliers and cooling designers to choose from. But once the design is done building it is possible and even though "*a heat reuse plant looks like an explosion in a spaghetti factory*" because of all the pipes, the consensus is that it is worth doing it. DC operator H also agrees that a data centre with heat reuse is more complicated from a technical point of view, and that having more components installed means having more things that can break. Moreover, the operation team needs to be trained on the heat recovery equipment as well. DC operator H thinks that some DC operators could avoid recovering excess heat due to this added complexity.

It is also worth mentioning that per definition the efficiency of a data centre with heat recovery and heat pumps is usually worse. DC operator H explained that efficiency is measured with the KPI called PUE (Power Usage Effectiveness) which is the total energy going in divided by the energy used by the IT equipment. The more the PUE is near 1 the more efficient a data centre is. Since heat pumps use electricity, the PUE is usually higher in a data centre which recovers its excess heat and upgrades it.

However, in the future, it could be possible to avoid using heat pumps. DC operator F mentioned that with other cooling methods such as immersion cooling, it could be possible to get rid of the heat pumps, but that is far in the future for mainstream applications. Also DC operator H talked about immersion cooling, which would mean having much higher temperatures of the excess heat from the start, around 50 °C or even higher. So, immersion cooling would offer better opportunities to heat recovery. DC operator H also mentioned how the fifth DH and cooling generation would enable higher potentials for using excess heat from data centres and even though the capital investments would be massive, there is a need to scale up and picture data centres as an integrated part of the energy systems.

- Perspectives of DH operators:

A technical issue preventing the use of more excess heat could be the fact that excess heat from data centres is low quality, which means that it should be upgraded with heat pumps to be used in a district heating network. According to DH operator A, that could be a barrier for DH operators which do not have heat pumps and an investment to recover the excess heat. DH operator B mentioned that they do not have heat pumps since their production is cheap, so heat pumps would be too expensive for them. However, DH operator B also stated that technically everything is possible more or less and that the main issue is usually having the proper business because that is more complex and technical problems are often easier to solve. A solution could be having the data centre introduce heat pumps in their cooling systems.

- Perspectives of municipalities:

The interviewee of Municipality C thinks that the main challenge in Stockholm is to have space to locate data centres near the city and district heating grids, especially considering big data centres which need a lot of space also for security reasons.

The SWOT matrix regarding the technical aspects can be seen in Figure 5.



Figure 5 - SWOT matrix for the technical aspects

4.2.3 Business aspects

- Perspectives of DC operators:

DC operator G stated that for district heating companies that already have a compensation model and a connectivity model, the process of implementing a heat recovery contract is easy. Those who do not would need to adapt their system, which could be a challenge. DC operator D also mentioned that the legal part of the heat recovery contract was an easy process. In Stockholm Data Parks, there already is a ready-made contract template to which typically not many changes are needed. Also DC operator H stated that not having standardised contracts and business models can be challenging, but it is not in Stockholm.

Regarding information security, all the DC operators interviewed agree that it is not an issue. In particular, DC operator D said that the only information they share is data on heat export and that they do not join any information systems. So, DC operator D thinks of their heat recovery contract just as a normal contract regarding information security. DC operator G mentioned that information security and heat recovery do not have anything to do with each other, considering how cooling systems are implemented, which are vastly decoupled from the IT load.

Regarding the business model itself, DC operator F said that they get paid per MWh that they supply and that the basic model is that they invest in heat pumps, take some extra electricity, supply heat, and get paid for it. The heat recovery contract lasts a couple of years and then it is possible to renew it, considering that the payback time might be longer than a couple of years. DC operator H said that every DC operator would want a long-term contract to recover the investments made for heat recovery. DC operator H stated that their income is based on the outdoor temperature, so they get a fixed compensation based on it per MWh.

- Perspectives of DH operators:

DH operator A stated that the business model is complex regarding what they should pay for the excess heat, as they cannot pay more than it costs for them to produce heat themselves; otherwise, it would be a loss. But at the same time, they would want to make it an attractive product for DC operators. Moreover, since DH operator A uses their heat pumps, the price of electricity has a huge impact. For example, DH operator A mentioned that it was a loss for them to buy excess heat when the electricity price was recently very high. However, it also depends on data centres. It is easier to make a business model which is also profitable for data centres when the latter only needs cooling and excess heat has no value. Apart from that, DH operator A does not think it is risky to make heat recovery contracts with DC operators since it does not happen very often that data centres shut down or move locations.

DH operator B stated that it could benefit DH operators if the DC operators pay for the heat pumps and install them since they would take a more significant risk. However, in that case, DC operators would want to get paid more for the excess heat, which could be better or worse depending on how the district heating grid looks.

- Perspectives of municipalities:

The interviewee of Municipality C emphasised how having a business incentive is essential for DC operators. To achieve such an incentive, a DH operator has to create a marketplace for waste heat. Once there is a marketplace established, other incentives or policies are not needed

to enhance more integration of waste heat from data centres. It's then up to the DC operator to understand if they can make a business with a DH operator.

The SWOT matrix regarding the business aspects can be seen in Figure 6.



Figure 6 - SWOT matrix for the business aspects

4.2.4 Environmental aspects

All the stakeholders interviewed agree that using excess heat from data centres in a district heating network has a positive environmental impact.

- Perspectives of DC operators:

DC operator D stated that they sell their excess heat because they want to do the right thing for society and that their internal environmental policy is to not waste energy.

Moreover, DC operator D also mentioned that more and more customers require heat reuse or strongly emphasise it. For example, almost all government branches in Sweden want to make deals with DC operators who have heat recovery. Thus, according to DC operator D, it is very valuable to have a partner (the DH operator) that can show that using their excess heat is positive for the environment since strict numbers on how much heat was exported and how it is used can be shown. Of course, DC operator D also does not want to lose potential customers. DC operator H thinks that 10 years from now, a DC operator will not be allowed to build a data centre without having evaluated if heat reuse is possible. That is what also thinks DC operator D, adding that new data centres being built in Sweden with no heat reuse should have as the only reason for not doing it, not being able to export the heat in any way.

DC operator F thinks that supplying their excess heat has a positive environmental impact, which, however, also depends on what sources are replaced and how the district heating grid looks. DC operator F also committed to only using electricity produced from hydro and wind farms every hour of the day with a solution developed by Vattenfall and Microsoft called “24/7 matching”.

DC operator E highlighted the importance of circular economy. DC operator F also mentioned that locating data centres near the city centres is important to integrate a circular economy in the sector and make data centres part of the energy system.

DC operator H mentioned that selling their excess heat could help reduce the peaks and even postpone or cancel the construction of new heat production plants.

- Perspectives of DH operators:

DH operator A said replacing something else that should be burnt always has a positive impact and that releasing excess heat into the surrounding area and thus wasting energy is not good. DH operator B stated that generally, it is always good to take advantage of the excess heat if possible and apply a circular economy in the sector.

- Perspective of municipalities:

The interviewee of Municipality C mentioned that using excess heat is important for saving resources. For example, more sectors compete over biomass which is used both as a heat source and to produce biofuels. It is thus important to be able to reduce the use of biomass in DH systems by using excess heat instead. Moreover, the positive impact could be more beneficial in DH systems that still have fossil fuels.
Finally, the interviewee of Municipality C added that DC operators are incentivised to establish their data centres in Sweden thanks to the high share of renewables in electricity production.

The SWOT matrix regarding the environmental aspects can be seen in Figure 7.

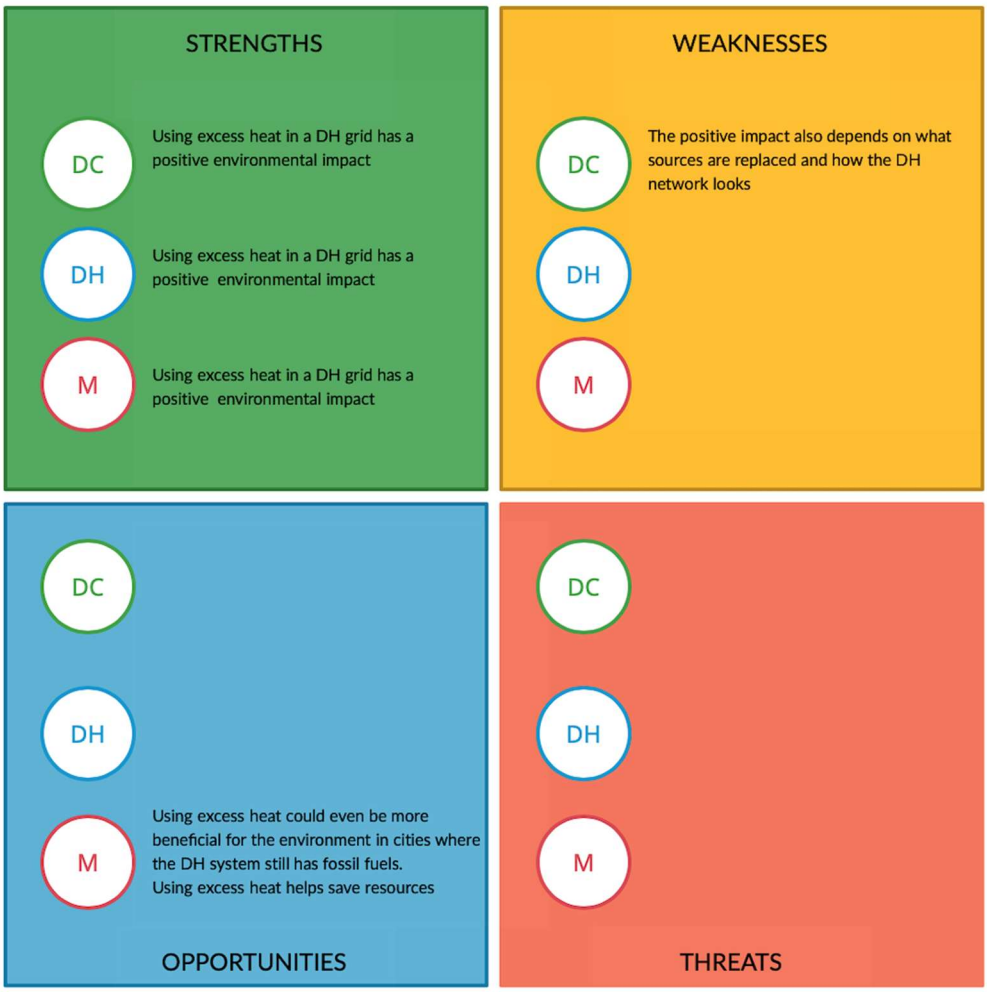


Figure 7 - SWOT matrix for the environmental aspects

4.3 Barriers

In Stockholm, heat recovery contracts between data centres and district heating companies seem to be well established mainly due to a strong social mindset pushing toward sustainability and a circular economy. The consensus of the stakeholders interviewed was that even though a heat recovery investment is not that good of a deal, it is the right thing to do.

The next sub-chapters focus on comparing the obtained data and the literature review. Four main aspects were identified as important.

4.3.1 Low-quality waste heat

The fact that waste heat from data centres is low quality was listed as the main barrier opposing to more heat recovery investments by Ebrahimi et al. (2014), Oró et al. (2019), Wahlroos et al. (2018) and European Commission (2020b). The possibility of this being a barrier in some contexts emerged also from the interviews done, as having low quality heat means that heat pumps need to be used. Having to install heat pumps could become a barrier because it is an additional investment that a DC operator might not be willing to make, considering that it can bring some technical concerns since it adds some complexity to the cooling system. Moreover, heat pumps increase the PUE since more electricity is needed. Thus, per definition the energy efficiency decreases. Wahlroos et al. (2017) mentioned that the PUE is actually overly simplified, but it is still the benchmark metric, thus many DC operators decide to not heat reuse. If the DH operator installs the heat pumps, this barrier could be less prominent depending on how the grid looks. Specifically, if the DH operator has to invest in heat pumps but has a cheap production, it will not be worth doing since heat pumps would be too expensive. However, for a heat recovery investment to be successful, it is essential to have a DH operator willing to buy the excess heat no matter how much DC operators would want to get rid of it. All the DC operators interviewed installed their own heat pumps because they were willing to take the challenge to do the right thing. In Stockholm, having low-quality waste heat does not generally seem to be a real barrier, considering that the payment for the waste heat compensates for the extra costs. Moreover, heat recovery gives DC operators a competitive factor since more and more customers demand heat recovery. Moreover, as emerged from some interviews, new technologies such as immersion cooling and low-temperature DH system will bring even more opportunities to recover waste heat from data centres in the future.

4.3.2 Lack of heat demand nearby

Wahlroos et al. (2018) mentioned that data centres should be near the DH network for the heat recovery investment to be feasible. Otherwise, the connections fees would be too high. DH operator A mentioned that getting the data centres near the city and their grids is their main challenge. When data centres are not close to their production facilities, the connection fees are too high and the pipe's dimensions could be too narrow. DH operator B said that they do not have data centres near their grids, so they would not be able to have waste heat from data centres in any case. The interviewee of Municipality C also specified that finding space for data centres near the city and grids is the main challenge in Stockholm. This could somewhat be a barrier in Stockholm as well for having more heat recovery investments. However, solutions like Stockholm Data Parks can potentially lessen this barrier.

4.3.3 High investment costs

According to Wahlroos et al. (2018), the high investment costs for the heat recovery equipment and heat pumps, which require electricity and add additional operation and maintenance costs, can be a barrier to waste heat utilisation. This also emerged from the DC operators interviewed, which agreed that a heat recovery investment is not that good of an economic deal. However, these DC operators feel a social responsibility to not waste energy and be as sustainable as possible, making them willing to invest. However, it is essential that the investment is feasible and that the payment for the waste heat covers the extra costs. High investment costs could surely put DC operators off in places with less environmental awareness and business models. Investment costs could be a barrier for DH operators as well. For example, DH operator A said that when the electricity price is too high, waste heat has no value for them because it would be too expensive to produce it into primary energy. How the grid looks and what production units are in it is also a factor when it comes to economic feasibility for DH operators.

Even though heat recovery contracts are not attractive economically, from a marketing perspective, they could be. Since sustainability is getting more and more important, DC operators who heat reuse could be able to expand their market.

4.3.4 Business and information security

Wahlroos et al. (2018) explained that since DC operators have competition in their markets and are privately owned, while DH is a natural monopoly and is partially publicly owned, there could be a lack of transparency from DC operators. This can reduce the possibility of commercialising excess heat due to information security. However, the DC operators interviewed agree that information security is not an issue when making heat recovery contracts. Moreover, there is a consensus among them that despite having a competitive factor with heat recovery, since sustainability is more and more requested from customers, they would want every DC operator to heat reuse. In RINA-C (2018), it is reported that information security and continuous operation have to be guaranteed. Indeed, some DC operators mentioned that they had to also invest in redundancy systems and local cooling systems to avoid possible disruptions, which could prevent some DC operators from selling their excess heat since the investment cost would get higher.

Wahlroos et al. (2018) also mentioned that business models for heat recovery investments are often missing or not clear. RINA-C (2018) argued that proper business models have to be applied to successfully implement heat recovery contracts. In Stockholm, since the business models are well established as in the marketplace, the business part of heat recovery contracts does not represent a barrier. However, considering what the interviewees mentioned, it seems like not having a proper business model in place could be the main barrier to more heat recovery. So, for having successful commercialisation of waste heat from data centres is essential to give DC operators a business incentive, as said by the interviewee of Municipality C, and have DH operators with established business models and standard contracts, which make the contract process straightforward.

4.4 Opportunities

The main opportunities in Stockholm revolve around sustainability. All the interviewees agree that using excess heat from data centres in a DH system always has a positive impact. In particular, it could help save other resources like biomass. Moreover, it can also replace pollutant heat sources such as fossil fuels. As a matter of fact, Wahlroos et al. (2017) mentioned that making use of waste heat is an environmentally sustainable practice since waste heat is an inevitable by-product.

As reported in RINA-C (2018), DC operators can increase their green image by selling their excess heat. Some of the DC operators mentioned that selling their excess heat can indeed give them a competitive factor marketing-wise. So, DC operators can contribute to making the energy system more sustainable while gaining competitiveness and potentially new customers.

In the literature, Wahlroos et al. (2017) stated that DH operators can lower their production costs by buying excess heat. DH operator A mentioned that being able to reduce their production costs represents a big opportunity for them.
Wahlroos et al. (2018) explained that DC operators who sell their excess heat can have additional revenue. Some DC operators interviewed mentioned that a heat recovery contract can be profitable in the long run.

Finally, some DC operators suggested that new technologies such as a low-temperature DH system could enable more opportunities of using low-quality excess heat. That is because heat pumps would not be needed. Wahlroos et al. (2018) also mentioned how a low-temperature DH system would facilitate the use of waste heat from data centres.

4.5 Social responsibility

Even in a city with an established marketplace for heat recovery investments, the economic attractiveness of such contracts is low. However, the results suggest that this possible barrier is overcome in Stockholm thanks to the stakeholders' involvement's environmental awareness. The DC operators stressed that heat reuse is the right thing to do and that they feel a social duty to do it. Thus, pushing toward a greener and more sustainable reality seems to be key in successfully developing heat recovery. This push is also felt by the customers of data centres who are now starting to emphasise heat recovery or even require it. So, the DC operators with heat recovery gain a competitive factor and could potentially attract more customers. Besides environmental awareness, DC operators also choose to make a heat recovery investment for marketing reasons. But it still all derivates from not wanting to waste energy that can still be used, even from the customers' perspective. Moreover, the interviewees seem to be aware of the importance of implementing a circular economy in the sector as the circular economy concept was brought up by some of them.

4.6 Stakeholders role – integrated perspective

The sustainability of data centres cannot be evaluated per se, but need to be holistically pictured. Data centres have to be integrated into the city energy system. Thus, it is significant that the stakeholders involved in heat recovery work together.

The results suggest that in Stockholm heat recovery investments revolve around mainly four factors:

1. DH operators with a good business model in place
2. Suitable space for data centres near the DH grids
3. Support from a municipality
4. Social responsibility

A good business model creates a marketplace and encourages DC operators to make a heat recovery contract. Suitable space near the DH grids is also essential since the connection fees would then be too costly for a DH operator. This could be somewhat troublesome, but initiatives like Stockholm Data Parks seem to be a good solution. That's where a municipality comes into play, as it can support such initiatives, and it can also help in the city planning to bring data centres closer. Finally, social responsibility is also an important factor because even with a proper business model and possible technical solutions, these investments are not that good of an economic deal. Moreover, pushing towards more sustainability and a circular economy for the good of society does not only affect DC operators but also DH operators, who have to be willing to accept the excess heat from data centres.

5 Conclusions

This study aimed to identify barriers and opportunities to heat recovery investments between data centres and district heating companies. It also aimed to understand how municipalities could enhance a greater integration of excess heat from data centres in a DH system. These purposes were achieved by developing a case study in Stockholm and interviewing the main stakeholders in this context: DH operators, DC operators, and municipalities. Insights were derived from conducting a SWOT analysis of the information collected from the interviews.

The study results suggest that the main barriers preventing heat recovery investments are mostly related to business rather than technical aspects. Even though in the literature, the main barrier is considered to be the low quality of the waste heat, which certainly adds challenges, possible technical solutions can still be reached. Moreover, new promising technologies such as immersion cooling and low-temperature DH system will bring more opportunities and technical innovation is always ongoing. A lack of a clear business model can make such investments challenging to implement since a marketplace can only be established with clear business models. However, DH operators with standardised models can help make the process of implementing a heat recovery investment easier. In Stockholm Data Parks there are ready-made contract templates to which usually many changes are not needed. The results suggest that DC operators are more willing to undertake a heat recovery investment if the DH operator takes care of the contract process.

As for opportunities, DC operators can earn additional revenue by making a heat recovery contract and DH operators can lower their production costs. Most importantly, a heat recovery contract adds sustainability, giving DC operators a competitive factor in their market. However, the results suggest that the main opportunities heat recovery investments bring are an improvement in sustainability performance and the possibility of implementing a circular economy in the sector.

5.1 Answering RQ1 and RQ2

It does not seem like there are significant barriers in Stockholm but rather challenges. However, some of these challenges could be barriers in other places. These possible barriers were derived using the SWOT analysis, considering the variables “weaknesses” and “threats”.

Regarding the economic aspects, the weaknesses and threats overcome strengths and opportunities because a heat recovery investment is not a good economic deal. If the DC operator has to buy and install the heat recovery equipment, additional costs are required, which a DC operator might not be willing to have. The investments needed can be important for several reasons discussed by the DC operators interviewed. Even though a profit can be made out of excess heat in the long run, from a merely economic point of view a heat recovery contract is not good for a DC operator. Thus some DC operators could avoid making a heat recovery contract for economic reasons. Moreover, a data centre with heat recovery is generally more expensive than a data centre without it. If the DH operator takes care of upgrading the waste heat quality, there might be some additional costs that would make the excess heat without any value. These additional costs include heat pumps and the electricity needed to run them. However, the feasibility for a DH operator is very location-oriented, since it depends on how the DH grid looks and what other sources are used. There might be competition with other heat sources that are less expensive for DH operators. Generally speaking, a DH operator cannot

pay the excess heat more than what it costs them to produce heat themselves; otherwise, it is a loss.

Considering the technical aspects, the main barrier is that the excess heat is low-quality, so heat pumps are needed to upgrade its quality. Having to install heat pumps introduces possible economic barriers and potential technical barriers. That is because designing the cooling system of a data centre is more complicated if the data centre recovers its excess heat. Moreover, retrofitting a data centre to make heat recovery possible could be challenging. A DC operator could then prefer to avoid this complexity and stick with a traditional cooling system, considering that the investment is not attractive economically. Moreover, the PUE increases with heat pumps as more electricity is needed. Some DC operators might want to avoid a worse PUE. Another possible technical barrier could be not having data centres near the DH grids due to a lack of suitable space.

However, the business part of a heat recovery investment could represent the most significant barrier since a well-established business model is necessary for having successful heat recovery contracts. That is mainly due to three reasons:

1. Proper business models in place create a marketplace
2. If the contract part is easy, DC operators may be more willing to make a heat recovery contract
3. For a DH operator without a business model or a not clear one, it might be challenging to accept and buy excess heat

In regards to the environmental aspects, there does not seem to be a possible barrier. Using waste heat instead of other sources can be more or less positive depending on what sources are replaced and how the DH grid looks but there is a general agreement among stakeholders that it is always good to not waste energy and reuse resources as much as possible.

Looking now at the variables “strengths” and “opportunities”, from an economic point of view, DC operators can still profit out of a heat recovery contract and thus have the opportunity for additional revenue. Instead, a DH operator could have a lower production cost by buying excess heat from data centres.

Considering the technical aspects, recovering waste heat makes a data centre more sustainable and gives a DC operator the opportunity of acquiring a green image. The main reason behind a heat recovery investment for the DC operators interviewed is that they can do the right thing for society by doing it. Focusing on Stockholm in particular, the opportunity to sell excess heat for data centres is ample since there is a marketplace and the DH network is one of the largest. Moreover, with new technologies such as immersion cooling and low-temperature DH system, there may be even more possibilities of using excess heat since it could be avoided using heat pumps.

As for the business aspects, when a DH operator has a proper business model in place, the contract process is easy, and thus, more DC operators could be willing to sell their excess heat. This enhances the opportunity of using waste heat from data centres in a DH system.

Regarding the environmental aspects, the stakeholders interviewed agree that using excess heat in a DH system always has a positive impact. In particular, it could help save other resources that do not have to be incinerated and it could be especially beneficial in the DH system where fossil fuels are still used.

5.2 Answering the RQ: *How can municipalities help to boost heat recovery from data centres?*

In the EU, policies targeting the sustainability of data centres are starting to be developed. However, the effort does not seem to be enough since, for example, both the EU Code of Conduct and the EU Green Public Procurement criteria for data centres are voluntary. Koronen et al. (2020) also conclude that there should be a more significant push to develop policies focusing on data centres since it is an important step to enhance waste heat recovery. However, the interviewee of Municipality C does not think that financial aid and policies are needed in Stockholm since there is already a marketplace, which is the most important incentive for DC operators. According to the interviewee of Municipality C, the main challenge is to find suitable locations for data centres near the city and the DH grids in order to be able to exploit the potential of using their excess heat.
Moreover, the need to attract data centres closer was also brought up by different stakeholders. Thus, what a municipality can do is help to find space for data centres, and one significant example is offered by Stockholm Data Parks, which is also supported by the city. When a DH operator is also co-owned by a municipality, the latter could favour the introduction of heat recovery contracts by pushing their DH utility to create a marketplace since it has political representation in the board. Lastly, municipalities could work on enhancing awareness towards sustainability and circular economy.

5.3 Contributions

This study has contributed to understanding the possible barriers and opportunities in heat recovery investments since there is a lack of best practices and investigations on the topic. In particular, the comparison between the reviewed literature and the obtained data aimed at understanding what applies in Stockholm and what does not. Finally, four main factors revolving around heat recovery investments in Stockholm were identified. These factors could help further heat recovery implementation in other locations.

5.4 Future research

Future research could focus on gathering more data from municipalities and DC operators not doing heat recovery, since it would be valuable to understand their perspective. Moreover, since SWOT lacks quantitative weighting, a further step could be a quantitative study.

More case studies in different locations would also be needed, as the implementation of heat recovery investments could be location-oriented.

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Appendix

Interview protocols

DH operators

Introduction

Does your company allow the use of excess heat recovered from third parties, such as data centres?

Economic aspects

What are the costs associated with heat recovery?
What price should the heat recovered from a data centre be in order to be selected over other sources, such as CHP plants?
Are there any economic benefits for your company when choosing heat recovered from a data centre? And for end users?

Technical aspects

Heat recovered from a data centre is usually low quality, is that a barrier for its commercialisation?
Do you think that a greater integration of low quality heat would speed up the development of a district heating network with lower distribution temperatures?
What is the optimal technical solution for recovering waste heat from a data centre?

Business aspects

Do you think it is risky for your company to make a heat recovery contract with a data centre?
Would you consider having a different business logic with data centres a challenge for making such contracts?
Do you think there is a need to have clearer business models? For example, it seems like often a not clear key question is who should pay for the heat recovery equipment.

Environmental aspects

Do you think that using excess heat from a data centre could have a positive environmental impact?

Wrap up

Can you think about specific challenges concerning the use of excess heat from a data centre in your company?
And opportunities?

DC operators, not in Stockholm Data Parks

Introduction

Does your company commercialise excess heat to district heating operators?

Economic aspects

What are the costs associated with heat recovery?
Is economic feasible for your company to commercialise excess heat?
If your company commercialise excess heat, does it have economic benefits through this commercialisation?

Technical aspects

What do you think is the optimal technical solution for recovering waste heat in a data centre?
Do you think that recovering waste heat would make a data centre more efficient and sustainable?
Would storing excess heat be feasible?

Business aspects

Do you consider having a different business logic with district heating operators a challenge for making heat recovery contracts?
Do you think there is a need to have clearer business models? For example, it seems like often a not clear key question is who should pay for the heat recovery equipment.
Does information security represent a barrier for your company to make heat recovery contracts?
Would your company be willing to make long-term contracts?

Environmental aspects

Do you think that commercialising your excess heat has a positive impact on the environment?

Wrap up

Can you think about specific challenges concerning the commercialisation of your excess heat to a district heating operator?
And opportunities?

DC operators, in Stockholm Data Parks

Introduction

Why did your company decide to build a data centre in one of Stockholm Data Parks?

Economic aspects

What were the costs associated with this initiative for your company?
What are the economic benefits for your company?

Technical aspects

From a technical point of view, do you think it is easier to build a data centre with already heat recovery equipment connected to a DH network instead of adding it later on?

Business aspects

Considering the different business logics DC and DH operators have, how was the contract process carried out?
What are the parts of this initiative the DH operator is taking care of?
How was information security handled?

Environmental aspects

Do you think that taking part in this initiative lessens the environmental impact of your company? If so, why?

Wrap up

What were the challenges concerned with this initiative?
What were the opportunities that made your company decide to build a data centre in Stockholm Data Parks?
Do you think that any data centre operator, even small ones, could build a data centre there?

Municipalities

Introduction

How much is your municipality aware of the possibility of using waste heat recovered from data centres in a district heating network?

How much is your municipality involved in your co-owned district heating network's business?

Economic aspects

What is the optimal financial aid when it comes to enhancing the integration of heat recovery from data centres into a district heating system? What about fiscal policies, like taxation?

Initiatives and policies

What type of initiatives and collaborations do you think can help boost heat recovery from data centres?

Do you think that policies targeting specifically heat recovery from data centres would sustain a better integration of it? If so, which kind of policies?

Environmental aspects

Do you think that using more excess heat recovered from data centres in a district heating network could positively impact the environment?

Wrap up

Can you think about the challenges your municipality faces when integrating more heat recovered from data centres into a district heating system?
And opportunities?